

Understanding the Challenges in Academia to Prepare Nursing Students for Digital Technology Use at Workplace

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ABSTRACT

Digital technology has transformed workplaces in recent years. The technology adoption in medical institutions has challenged nurses to evolve in using complex digital systems. However, academic institutions often fail to keep up with the rapid changes in professional practice in preparing their students. To this end, we conducted semi-structured interviews with five nursing faculties in the USA. In our study, we examined the current status of digital technology and privacy education, and identified the challenges in adequately preparing nursing students to embrace and adapt to the changes in digital technology use at healthcare organizations. Overall, our findings provide valuable insights for the CSCW community to better understand the challenges in academia, in order to mitigate the gap between academic preparation and professional needs of nurses in using digital technologies. Based on our findings, we provide recommendations to address these issues.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **User studies**; • **Security and privacy** → **Human and societal aspects of security and privacy**.

KEYWORDS

Nursing; needs in technology use; challenges in academic preparation; qualitative study

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1 INTRODUCTION

The rapid rise of digital technology and applications have transformed a majority of sectors [24, 33]. While professional organizations have rapidly adopted newer technologies to reduce costs and mistakes, and so on, improve productivity and performance, the academic institutions often fail to keep up with technological advancements. That can lead to challenges in appropriately preparing the students to use digital technology in their future practices. In this study, we start to explore these challenges in the context of nursing professionals.

The nurses play a critical role in patient care, and are the primary healthcare workers responsible for a wide range of tasks, including collecting, storing, retrieving, and updating patients' sensitive health information [13, 16]. The patient's medical record accumulates significant personal information, including identification, genetic information, psychological profile, history of medical diagnosis and treatments, sexual preferences, employment history, and income [22, 25]. With the recent adoption of technology in the healthcare sector, these health records are now maintained in digital form, resulting in nurses being tasked to adapt to new technology use [6, 9, 10, 15, 26]. The prior studies [30, 35–37] highlighted the challenges nurses face in using digital technologies at their workplace. These challenges can lead to mishaps with severe consequences, including the leakage of sensitive health information [7, 21, 27, 29, 31]. To this end, academia should be the starting point to train future nurses in handling sensitive information while using digital technologies at workplace, indicating the need of academia to better prepare their nursing students to adapt to technology use in their professional field [30].

A body of work [14, 18, 23, 34] investigated technology-enhanced learning experiences in academia for nursing students. Webb et al. [34] identified the benefits of using information communication technology in nursing education, where students demonstrate positive attitudes toward online and electronic learning media. The authors have suggested that administration support needs to be in place to enable greater adoption of information communication technology in pre-registration nursing [34]. Johnston et al. [23] explored the use of e-learning and digital gaming technology in nursing education and suggested a need for more critical engagement in the academic field of technology-enhanced learning [23]. In a separate study [14], Burke emphasized the stress nursing faculties face during course preparation and delivery using technology.

PID	Gender	Age Range	Race	Education
P1	Female	55-59 years old	White	Graduate degree
P2	Female	30-34 years old	White	Graduate degree
P3	Female	40-44 years old	White	Graduate degree
P4	Female	60-64 years old	White	Graduate degree
P5	Male	45-49 years old	White	Graduate degree

Table 1: Demographic Information of Nursing Faculty

The study of Duke et al. [18] reported on the use of social media by nursing students and faculties in education. The authors found that YouTube and text messaging are popular platforms for educational purposes [18], where more students than faculties reported that they were aware of privacy features and of the professional behavior expected when using social media. The authors noted the need for more privacy education around technology use in academia [18].

However, a little study to date, focused on identifying the challenges in adequately preparing nursing students to use digital technologies in their professional practice. We addressed this gap in our work. In particular, we focused on the viewpoint of academia (lacking in prior works), where we interviewed five nursing faculties who taught at least one course related to digital technology in nursing schools. Our key contributions are two-fold: i) Understanding the status of digital technology education in nursing schools (see §3.1), and ii) Identifying the gaps between academic preparation and professional needs of nurses in digital technology use (see §3.2). Our findings lead to the recommendations for mitigating these gaps (see §4.1), where we also provide the guideline for future research in these directions.

2 METHODOLOGY

We conducted semi-structured interviews with five nursing faculties in the USA. In preparing and refining interview questionnaires, the authors of this paper conducted focus-group discussions. The interviews were conducted over Zoom and took between 30 and 60 minutes. The studies were approved by the Institutional Review Board at our university.

Participants. To recruit participants, we sent invitation emails to the nursing department at several universities and used snowball sampling. We specified that any faculty members in the nursing department who taught courses on digital systems could take part in our study. During our interviews with the faculties, we learned that a minimal number of courses on digital systems are offered at nursing schools, which in many cases, are taught by a single faculty member in the department. We speculate that, due to these reasons, we heard only from a small number of participants. Table 1 reports the demographic information of our participants.

Procedure. When a participant showed interest in participating in our study, we emailed them the informed consent document (ICD). As they agreed to ICD, we scheduled a time for an online interview over Zoom. Interviews were audio recorded. During the study, we asked participants about the current status of digital technology education in their department, their perspectives on the gaps between academic preparation and the professional needs

of nurses, factors contributing to these gaps, and how we could bridge them. At the end of the interview, participants completed an online survey on demographics. They were compensated with a \$25 Amazon.com gift card.

Analysis. The audio recordings from the interviews were transcribed, and we then performed thematic analysis on our transcriptions [5, 8, 11, 12]. Two independent researchers coded each transcript, where they read through the transcripts of the first few interviews, developed codes, compared them, and then iterated again with more interviews. After all the interviews had been coded, 89.40% of the codes matched between the two reviewers. Cohen's Kappa score was 0.84, indicating good agreement between the coders. We discussed and reached an agreement regarding the discrepancies in the remaining codes. Finally, we organized and taxonomized our codes into higher-level categories.

3 RESULTS

In this section, we present the findings from our user study with nursing faculties. For consistency, we use these terms based on the frequency of comments in participants' responses: a few (0-20%), some (21-40%), about half (41-60%), most (61-80%), and almost all (81-100%).

3.1 Status of Technology Education in Nursing Schools

Most participants discussed the shift towards electronic health records from paper charting in healthcare organizations. One of the faculties (P2) who also worked as a nursing professional mentioned, *"I have been a nurse for a little over eight years now. When I first started as a nurse, there was much emphasis on paper documentation and not many digital resources. At my first job, they had just switched to an electronic health record system, and we were trying to teach and adjust to what we wanted. There was a big learning curve, especially for those who had been nurses for a long time and tried to switch from paper documentation to this system."* With such recent changes in using digital technologies at healthcare organizations, in this section, we present the current status of digital technology and privacy education at nursing schools in the USA.

Digital Technology Education. Most of our participants talked about academic courses related to electronic health records. One of them (P2) reported, *"We have one class in particular that talks about informatics in healthcare that focuses on many different technologies. When we talk about digital technology, our primary focus is on electronic health records and how to make the most of them, use them, and document them effectively."* One participant (P3) mentioned medicine scans and wearable technology (e.g., Apple Health with Apple Watch) as the areas she covered in her course. In their courses, about half of our participants taught topics related to big data in educating nursing students about digital technologies. One nursing faculty (P5) said, *"We teach them about big data, how it comes in, and why it is important to have all these synced data."* Most nursing faculties believe adopting digital technology in nursing schools benefits their students. One of them (P5) expressed his excitement in this context: *"When I get to see students in person, they talk about cool ideas that use digital technology like a stethoscope*

that could record sounds and send it somewhere, which helps me see that they are thinking about technology with healthcare and how to push them forward for improvement."

Digital Privacy Education. Almost all participants reported that privacy in digital healthcare technology is covered in nursing schools, where the central focus is on Health Insurance Portability and Accountability Act (HIPAA) ¹. One faculty (P4) said, "...that [HIPAA] is a major concept and theme that goes throughout our curriculum. So, we started teaching them about confidentiality very early on about HIPAA, signed HIPAA and confidentiality agreements with the college and each clinical site they attend." While HIPAA is covered in their course, P2 reported the lack of information provided to nursing students about keeping digital information safe; she said, "As far as just general privacy... we do not spend much time on how to keep digital information safe as we found that depending on where they end up for practice, each facility has its tech on how to keep information safe, how to avoid phishing scams, how to avoid losing password, keep actual digital files, digital information safe."

3.2 Challenges in Academia: Making Sense of the Gap

Keeping up with Technological Changes. We identified two primary reasons: financial barriers and fast-paced changes, as we looked into why it is challenging for academia to keep up with educating students on technology use with the advancement in technology at healthcare organizations. In these contexts, P1 reported, "If they [medical organizations] get a new fancy machine, we cannot duplicate everything they get at the hospital. We can only show videos [to students] in this case. We teach them all the basic skills, but some machines are not affordable to teach in the school." While finances result in difficulties in acquiring new devices for educational purposes, some participants pointed to the fast-paced updates of medical software, posing challenges for them as requiring frequent changes in course contents and learning the updated software before teaching their students.

Integration of Privacy Education. Our participants highlighted the difficulty in integrating privacy into nursing education, where they pointed to the confusing legality associated with privacy documents. One of them (P5) reported, "I think it is confusing with all the legal wording and meanings of acronym making it hard to remember properly." They also referred to the difficulty in effectively conveying the criticality of privacy within a limited time, where P4 said, "So I think that can be difficult. And then I think in our community, it is really about remembering that even though this person is your cousin's cousin, you were not their nurse, so you cannot talk about what you know with other people. And that concept, I think, has to be pretty hard." In a similar context, P3 mentioned, "Possibly the hardest part is quickly helping students understand the level of importance."

Collaboration with Medical Organizations. Almost all participants pointed to a need for collaboration between medical institutions and nursing schools; P3 reported, "Yes, I think most institutions can work with students to help reduce the gap to some degree. However, there

are still many places where our students cannot chart because they are not nurses yet." Some faculties expressed frustration that the responsibility of preparing the students falls largely on academia, where they believe collaboration with medical institutions would benefit medical schools. Here, the lack of collaboration results in fewer hands-on experiences for nursing students, where P2 mentioned, "When our students go into Capstone in the last semester ... they are consistently coming back frustrated, feeling that they were not meeting the standard, thereby making them look bad or just struggling in general as they did not have exposure to those [digital] systems ... They all feel this will hinder their progress when they graduate...students feel underprepared in this area." While bringing new and updated technology to academia is challenging, collaboration with medical organizations can help academia bridge that gap.

3.3 Mitigation Strategies: Preparing Future Nurses

Technology-directed Focus. Almost all participants believe that adding more technological aspects to the courses being taught in the nursing schools is essential, where P5 said, "The assignments given in certain courses could include more technology-type things they [students] will see in the real world." Some participants also talked about conveying the importance of digital technology in modern nursing. One of them (P3) reported, "I think that assisting students in seeing and understanding how technology might make their lives easier and their jobs more efficient is important. ... So helping them to become excited about it and recognize all of the positive things is like how I would probably try to face that gap [difficulty with technology use] in class."

Introducing Perspectives of Nursing Professionals. Some participants pointed to the importance of bringing the experiences and perspectives of a professional nurse to the students. One of them (P4) said, "A clinical teacher (nurse) who is an insider or an employee of a system will be able to provide students with far more in-depth access to that system." Participants also mentioned that nursing professionals could share their experiences and convey the importance of digital technology to increase the student's interest in exploring it as part of their nursing education. One participant (P3) reported, "I could probably bring in more enthusiastic experienced nurses from outside our class as guest speakers to help the students understand that this [using digital technology in nursing] isn't something you learn when you're growing up and you're in high school."

4 DISCUSSION

4.1 Padding Gaps in Academia

Our findings shed light on the challenges faced by academic institutions in preparing nursing students to use digital technology in professional practice, leading to recommendations for addressing these issues, as discussed in this section, along with the avenues for future work in these directions.

Concept-based Education. Nursing students should understand the importance and general concepts around digital information protection, where our findings point to the gap in educating them about keeping digital information safe (see §3.1). While we found evidence that the lack of updated software creates hindrances in

¹<https://www.cdc.gov/php/publications/topic/hipaa.html>

educating nursing students about the latest technologies used in healthcare, we believe there is a partial but potentially effective solution to these issues. The software designed for the same purpose typically has similarities in core functionalities, although there may be variations in features and user interfaces. We speculate that learning the core concept of privacy-enhancing applications (e.g., VPN) would build a basic understanding among nursing students that they can later translate into using an updated or a different version of these applications at their workplace. We encourage future studies to explore the challenges in knowledge transfer regarding using a different or updated version of an application by nursing professionals.

De-Jargoning the Legalities. We acknowledge that the legal terms in privacy laws can confuse nursing students (see §3.2). However, courses in business and law schools use scenarios to de-jargonize the legalities². In such an approach, students are given scenarios as assignments and asked what actions to perform. For instance, a scenario can entail: “A nurse’s cousin is admitted to a hospital where she works on. The nurse is not assigned to her cousin, but she is very interested in her cousin’s condition. What should she do? What would be the consequences of looking at her cousin’s records?” Using such a scenario to explain the legalities would simplify jargon in legal documents and prepare nursing students for various situations they might encounter during their professional practice.

Adoption of Nurse Residency. Our findings point to the challenges in academia to keep up with technological changes in professional practices (see §3.2) and in providing hands-on experiences to nursing students on technology use (see §3.2). The clinical hours provide opportunities for nursing students to have hands-on interactions with digital technology at a healthcare organization [30], where our participants pointed out that medical organizations are hesitant to accommodate students due to liability concerns. Nurse residency program [4, 17, 19] could provide a potential solution in this regard, where the nursing graduates gain hands-on experience before stepping into their professional careers. While nurses get limited mentoring with technology use upon starting their professional career [30], the early-career mentoring program like nurse residency could help to bridge the gap between academic preparation and the professional needs of nurses. We encourage the wide-scale adoption of nurse residency programs at healthcare organizations to provide a learning opportunity to nursing graduates regarding digital technology use at the workplace. In these contexts, academic institutions could also leverage the benefits by inviting their recent graduates to share their experiences on digital technology use with nursing students.

4.2 Limitations and Future Work

We interviewed five participants in our study, which followed a qualitative research approach [8, 11, 12]. We acknowledge that a different set of samples might yield varying results. Thus, we do not draw any quantitative, generalizable conclusion from our study. While we looked into the challenges leading to the gap between academic preparation and the professional needs of nurses in technology use, we note that such gaps and challenges may also exist in

other fields of education, especially those impacted by the meteoric rise of technology. Therefore, we encourage future studies to focus on understanding the gap in technology use between academia and professional practice in different fields of education and develop contextualized solutions to bridge these gaps. Our study was limited to the U.S. participants only. The recent studies [1–3, 20, 28, 32] highlighted the importance of looking beyond the Western contexts. In addition, the gap between academia and practice may be further aggravated for international students in the USA due to differences in their academic preparation. We encourage future studies to look beyond the Western contexts and focus on the needs and expectations of nursing students from different backgrounds.

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²<https://myassignmenthelp.com/free-samples/education-law-scenario>

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